

Homoscedasticity and Its Testing: From Conditional Expectation to the Breusch–Pagan Test

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Part 6 / 12

**Three Asymptotically Equivalent Tests:
Wald, LR, and Score**

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1 The Three Likelihood Tests: Wald, LR, and Score

1.1 Drawing the log-likelihood: unimodal case

We begin by drawing the log-likelihood $\ell(\theta)$ as a function of the parameter θ , with data x held fixed. In most problems ℓ can be negative (always so for discrete distributions; not necessarily for Lebesgue densities), but the sign does not affect the geometry. We take the unimodal case.

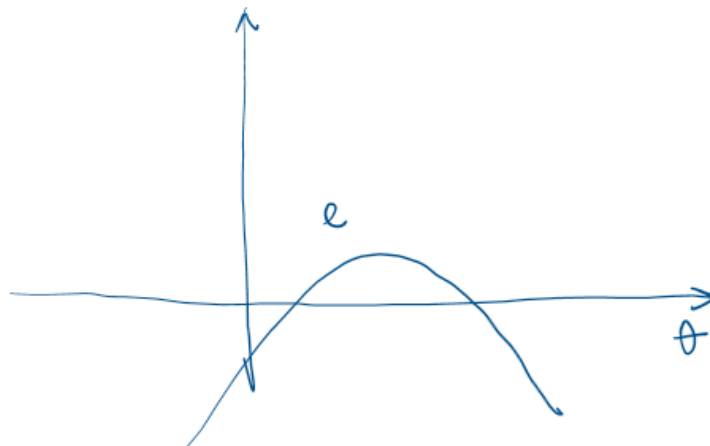


Figure 1: The Probabilist's sketch (2 June 2026): unimodal log-likelihood $\ell(\theta)$ as a function of θ , data x fixed. The curve has a unique maximum — the MLE.

The data x was drawn from a distribution with true parameter θ_0 . We mark θ_0 (“true”) and $\hat{\theta}_{\text{MLE}}(x)$ on the same graph. For this particular sample, the MLE falls to the left of θ_0 .

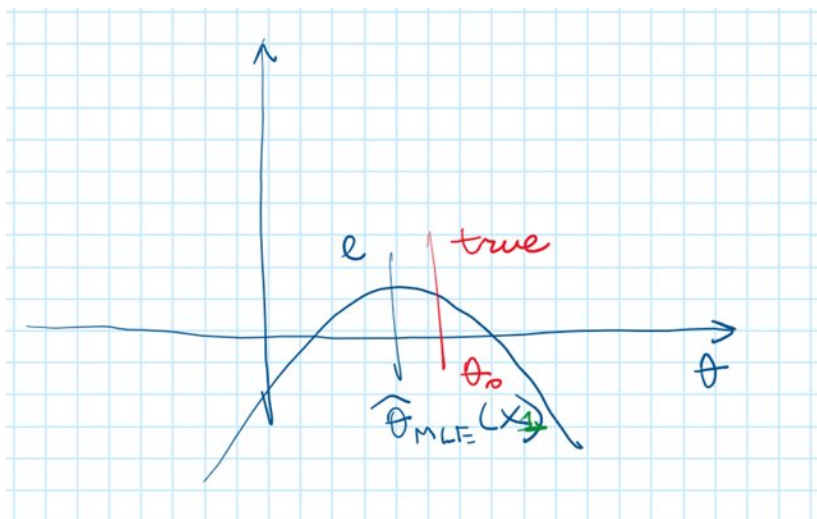


Figure 2: The Probabilist's sketch (2 June 2026): log-likelihood with θ_0 (true, red) and $\hat{\theta}_{\text{MLE}}(x_1)$ (green) marked. The MLE is sample-dependent.

1.2 The MLE is a random variable: two samples

Collecting a second independent sample x_2 from the same distribution P_{θ_0} yields a different log-likelihood and a different MLE. The two MLEs land on opposite sides of θ_0 .

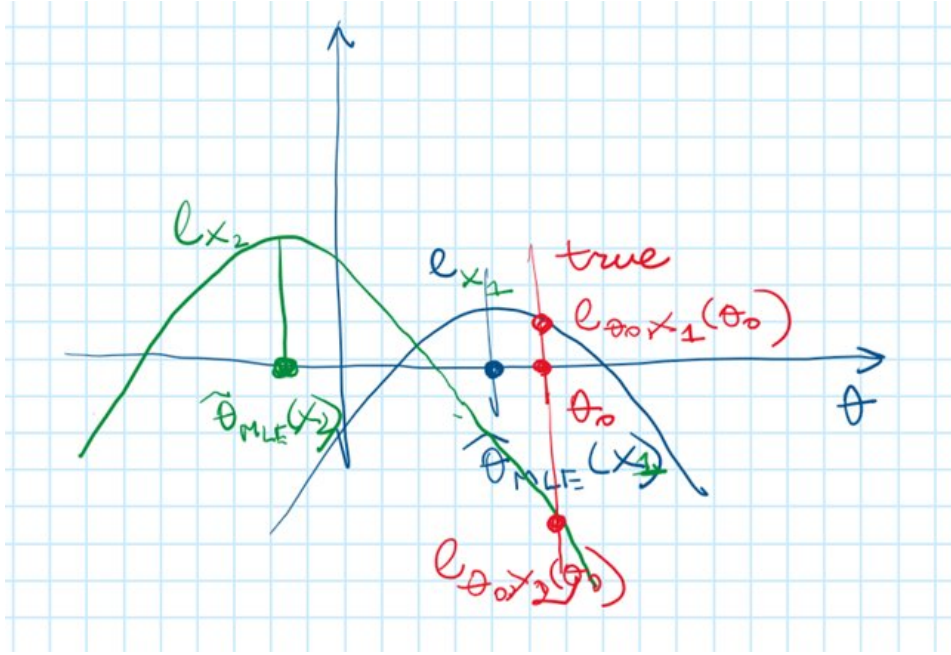


Figure 3: The Probabilist's sketch (2 June 2026): two log-likelihoods ℓ_{x_1} (blue) and ℓ_{x_2} (green) for two independent samples from P_{θ_0} . $\hat{\theta}_{\text{MLE}}(x_1)$ and $\hat{\theta}_{\text{MLE}}(x_2)$ land on opposite sides of the true θ_0 (red vertical). The red dot marks $\ell_{\theta_0, x_1}(\theta_0)$, the log-likelihood evaluated at the true parameter. The MLE is a random variable; the CLT describes its distribution around θ_0 .

There are **three approaches** to measuring the distance between the true θ_0 and the estimated θ . The estimate does not have to be MLE in general — but it is in the standard asymptotic theory. The following picture, from Engle (1984), shows the case where $\hat{\theta}$ is the MLE.

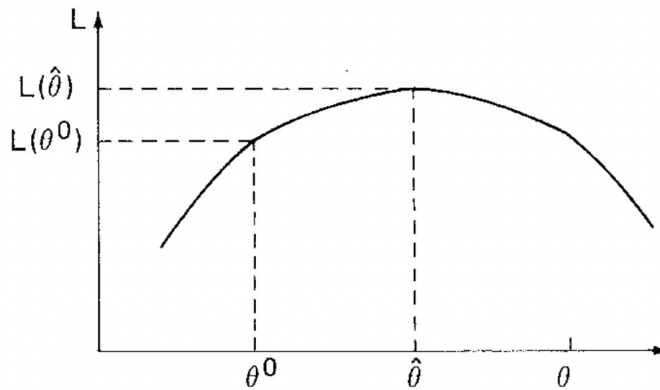


Figure 4: From Engle (1984, Ch. 13, Fig. 3.1): log-likelihood $L(\theta)$ with $L(\hat{\theta})$ at the peak and $L(\theta^0)$ at the null. The two dashed horizontal lines mark the two likelihood values; the two dashed vertical lines mark θ^0 and $\hat{\theta}$.

1.3 Three distances: formulas and geometry

Figure 5 shows all three distances simultaneously on the small- ℓ log-likelihood graph, together with the key labels θ_0 , $\hat{\theta}$ (MLE), and θ_{true} .

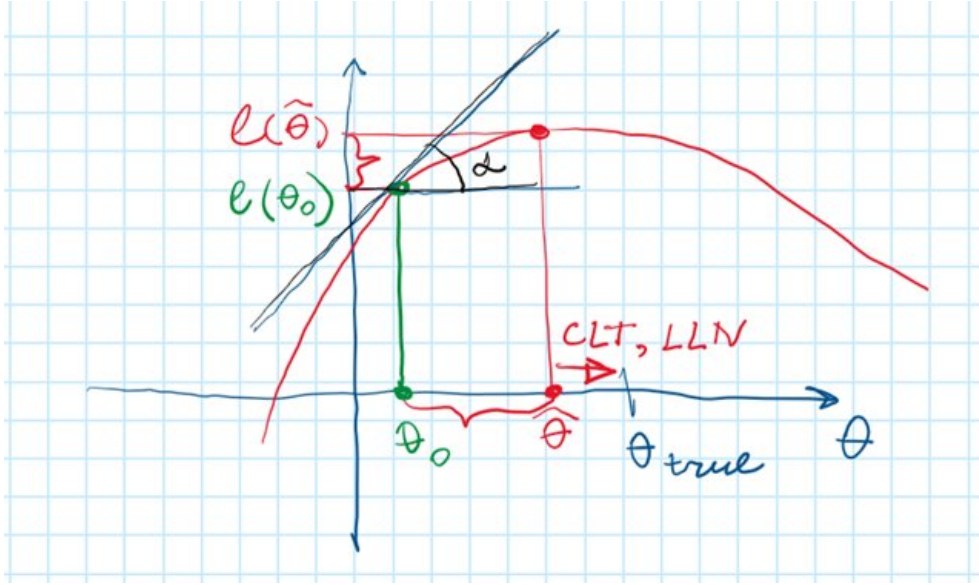


Figure 5: The Probabilist’s sketch (3 June 2026): the log-likelihood ℓ with θ_0 , $\hat{\theta}$, and θ_{true} on the horizontal axis. The three test statistics correspond to: **(1) Wald** — horizontal distance $\hat{\theta} - \theta_0$ (green bracket on θ -axis); **(2) LR** — vertical distance $\ell(\hat{\theta}) - \ell(\theta_0)$ (red bracket on ℓ -axis); **(3) Score/LM** — slope $\tan \alpha = \ell'(\theta_0)$ at θ_0 (the two tangent lines through $(\theta_0, \ell(\theta_0))$). Annotation “CLT, LLN” marks the convergence $\hat{\theta} \xrightarrow{P} \theta_{\text{true}}$. In all three distances we would prefer θ_{true} instead of $\hat{\theta}$, but we do not have it. Under H_0 : $\theta_{\text{true}} = \theta_0$, and then all three CLTs work.

On the picture we can consider two of the three distances directly:

$$d_1 = \hat{\theta} - \theta_0 \quad (\text{horizontal, Wald}), \quad d_2 = \ell(\hat{\theta}) - \ell(\theta_0) \quad (\text{vertical, LR}).$$

For the LR distance, note that the difference of small- ℓ log-likelihoods equals the logarithm of the ratio of the capital- L likelihoods:

$$\ell(\hat{\theta}) - \ell(\theta_0) = \ln L(\hat{\theta}) - \ln L(\theta_0) = \ln \frac{L(\hat{\theta})}{L(\theta_0)},$$

which explains the name *likelihood ratio* test. The third distance is the distance of the *score* from zero: $d_3 = s(\theta_0) = \ell'(\theta_0)$, the slope (“tan α ”) of ℓ at θ_0 .

Wald test. Estimates θ *without* the restriction (H_1 model) and measures how far $\hat{\theta}$ is from θ_0 in parameter space, scaled by estimated precision:

$$\xi_W = (\hat{\theta} - \theta_0)^2 \cdot \mathcal{I}_n(\theta_0) \xrightarrow{d} \chi_1^2 \quad \text{under } H_0.$$

Starts at the alternative. Requires fitting the unrestricted model.

Likelihood Ratio test. Compares both models by their log-likelihood values:

$$\xi_{LR} = 2[\ell(\hat{\theta}) - \ell(\theta_0)] = 2 \ln \frac{L(\hat{\theta})}{L(\theta_0)} \xrightarrow{d} \chi_1^2 \quad \text{under } H_0.$$

Compares both hypotheses directly. Requires fitting both models.

Score (LM) test. Estimates θ *only* under H_0 ; checks whether $\ell'(\theta_0)$ is near zero:

$$\xi_{LM} = \frac{[\ell'(\theta_0)]^2}{\mathcal{I}_n(\theta_0)} = \frac{s^2(\theta_0)}{\mathcal{I}_n(\theta_0)} \xrightarrow{d} \chi_1^2 \quad \text{under } H_0.$$

Starts at the null. Does *not* require computing $\hat{\theta}$ — the only test of the three without the MLE.

The third distance can be interpreted as an elasticity of L capital (it does not have as easy a graphical interpretation in the L picture), but on the small- ℓ graph the slope at θ_0 is visible directly.

Remark 1 (Why θ_{true} , not $\hat{\theta}$). In all three distances we would ideally use θ_{true} instead of $\hat{\theta}$. But θ_{true} is unknown. Under H_0 we identify $\theta_{\text{true}} = \theta_0$, and under this identification the distributions of all three statistics depend only on known quantities. This is what makes the CLTs work and all three converge to χ^2 .

Multivariate generalisation. With r restrictions on $\theta \in \mathbb{R}^p$, all three statistics share the χ_r^2 null distribution:

$$\begin{aligned}\xi_W &= T(\hat{\theta} - \theta_0)' \mathcal{I}(\hat{\theta})(\hat{\theta} - \theta_0), \\ \xi_{LR} &= 2[\ell(\hat{\theta}) - \ell(\tilde{\theta})], \\ \xi_{LM} &= \mathbf{s}(\tilde{\theta})' \mathcal{I}(\tilde{\theta})^{-1} \mathbf{s}(\tilde{\theta}) / T,\end{aligned}$$

where $\tilde{\theta}$ is the restricted MLE (under H_0). For a one-dimensional θ the score has a normal CLT limit; for the multivariate case we need to collapse to one dimension, hence the chi-squared.

1.4 Asymptotic equivalence and the quadratic likelihood

When ℓ is well approximated by a quadratic (which holds for local alternatives as $n \rightarrow \infty$), all three tests are identical. If $\ell(\theta) = b - \frac{1}{2}(\theta - \hat{\theta})A(\theta - \hat{\theta})$ exactly, then straightforward algebra gives $\xi_W = \xi_{LR} = \xi_{LM} = (\theta_0 - \hat{\theta})^2 A$. The asymptotic equivalence follows because the log-likelihood is asymptotically quadratic in a $1/\sqrt{n}$ neighbourhood of $\hat{\theta}$ (Engle 1984, Theorem 1).

Remark 2 (Why they differ in finite samples). The three statistics use different variance estimates: ξ_W uses $\mathcal{I}(\hat{\theta})$ (under H_1), ξ_{LM} uses $\mathcal{I}(\tilde{\theta})$ (under H_0), and ξ_{LR} uses both implicitly. For finite n this produces $\xi_W \geq \xi_{LR} \geq \xi_{LM}$ (Breusch 1979; Savin 1976). Consequence: whenever LM rejects, so do the others; whenever W fails to reject, so do the others. When sizes are corrected to be equal the power orderings flatten out.

1.5 Computational advantage of the LM test

The key practical reason to prefer the LM test is computational: *only the restricted model needs to be estimated*. This matters when:

- the restricted model is simple (e.g. OLS) but the unrestricted model requires iterative non-linear maximisation;
- one wants a battery of diagnostic tests against many alternatives — each LM test reuses the same restricted estimates;
- the null has already been estimated and the question is whether the data signal a departure.

For the Breusch–Pagan heteroscedasticity test, the null is OLS, and the LM statistic reduces to nR^2 of an auxiliary regression of squared residuals on x — no heteroscedastic model is fitted.

1.6 The score as a push-forward map

Figure 6 shows the score $s(x, \theta_0)$ as a function of x , overlaid on the density of P_{θ_0} .

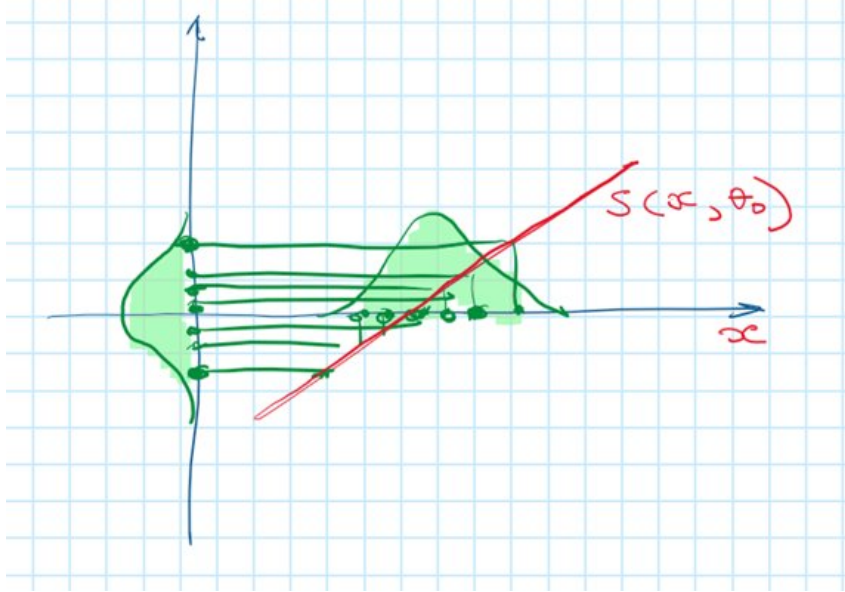


Figure 6: The Probabilist’s sketch (4 June 2026): score $s(x, \theta_0)$ (red line, increasing in x) overlaid on the density of P_{θ_0} (green area, left bell) and on the density of the pushed-forward image (right green area). When the sampling measure is “native” ($P = P_{\theta_0}$), the push-forward of P_{θ_0} under $x \mapsto s(x, \theta_0)$ is centred at zero. This is the identity $\mathbb{E}_{\theta_0}[s(\theta_0, X)] = 0$. In the unimodal case, when zero is near the mode of the score, the score test works best: the score is approximately a pivotal quantity whose distribution does not depend on μ .

Think of $x \mapsto s(x, \theta_0)$ as a *centering map*: it pushes P_{θ_0} to a centred distribution (mean zero). When the data instead come from P_{θ_1} with $\theta_1 \neq \theta_0$, the same map is applied to the “wrong” measure. The push-forward of P_{θ_1} will not be centred, and this off-centring is what the score test detects.

Figure 7 shows this disagreement explicitly.

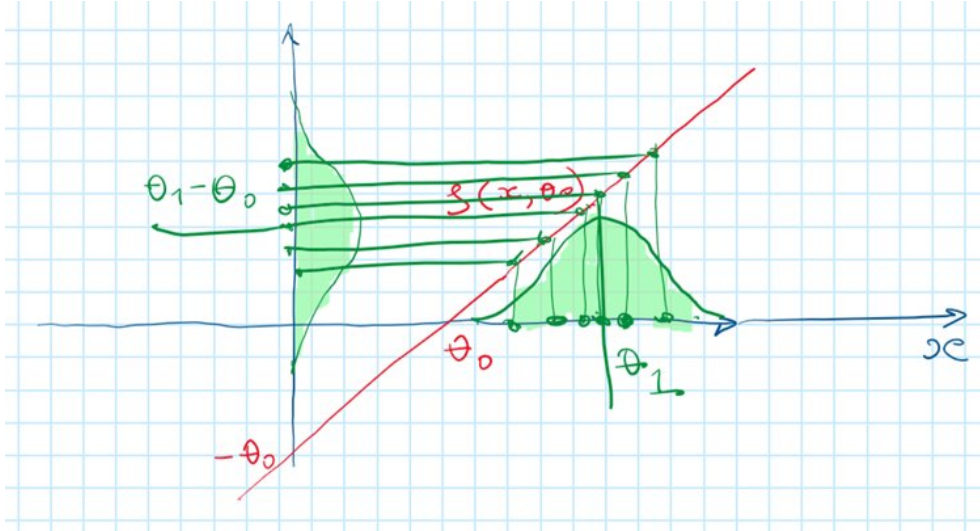


Figure 7: The Probabilist’s sketch (5 June 2026): score of the family $\mathcal{N}(\theta, 1)$, section at $\theta = \theta_0$ (the red line $s(x, \theta_0) = x - \theta_0$), overlaid on the density of $\mathcal{N}(\theta_1, 1)$ with $\theta_1 > \theta_0$ (green bell curve centred at θ_1). The green area is shifted to the right of θ_0 : the push-forward of P_{θ_1} under $s(\cdot, \theta_0)$ has positive mean $\theta_1 - \theta_0$. The label $\theta_1 - \theta_0$ on the vertical axis shows the shift of the push-forward measure away from zero. The score test rejects H_0 when the sample mean of $s(x_i, \theta_0)$ is far from zero relative to its standard deviation.

Remark 3 (Two identities connected to zero). The score is tied to zero by two distinct identities which are easy to confuse:

- (i) **Expectation identity** (under the true θ): $\mathbb{E}_\theta[s(\theta, X)] = 0$ for all θ . This is the LLN/CLT basis for the score test.
- (ii) **First-order condition at the MLE**: $s(\hat{\theta}, x) = 0$ for the observed sample. This is a sample identity, not an expectation.

The score test uses identity (i) with $\theta = \theta_0$: under H_0 the sample mean $\bar{s}(\theta_0)$ should be near zero (CLT). Identity (ii) is not used: $\hat{\theta}$ does not appear in ξ_{LM} .

The Probabilist’s observation: we do reject H_0 when the mapped sample is far from the central position — but “far” means the *average* score $\bar{s}$ is far from zero (via CLT for identity (i)), not that any individual $s(x_i, \theta_0)$ is large.